

Storyverse — Build Roadmap

Chapter-Accurate, Theatrically Immersive Character AI for Fiction

Reference Implementation: Harry Potter (Books 1-7)

Phase 0 — Fix What's Broken

Goal: Make the existing prototype airtight before adding anything new. **Timeline:** 3-4 days

0.1 Action Beat Diversity System

Problem: Characters default to 3-4 generic gestures (Hermione picks up quill, Ron laughs, Harry smiles). Feels robotic after 5 turns.

Solution:

- Build a per-character physical vocabulary mapped to emotional states — 10-12 unique actions per character
- Happy Ron ≠ nervous Ron ≠ angry Ron — each state triggers different physical behaviour
- Add hard rule: "Never repeat the same action beat within a conversation"
- Character profiles expand from 6 fields to 7: personality, speech, emotional triggers, excitement triggers, discomfort triggers, private context, **physical vocabulary**

Characters to build vocabularies for (13): Harry, Ron, Hermione, Hagrid, Dumbledore, Snape, Draco, Neville, Dobby, Ginny, Tom Riddle, Moaning Myrtle, Lockhart

Deliverable: Updated character profiles with physical vocabulary field; updated prompt template enforcing no-repeat rule.

0.2 Session Persistence

Problem: Conversation history lives in JS memory only. Page refresh, navigation, or API errors wipe it. Users lose context after a few messages.

Solution:

- Write `(hist)` to persistent storage after every turn, keyed by `(bookId + chapIdx + charId)`
- On character load, check for existing history and restore it with a subtle "Continuing where you left off..." indicator

- Add a "New Conversation" button so users can intentionally reset
- Handle edge case: if user changes chapter but keeps the same character, clear history (knowledge boundary changed)

Deliverable: Conversations survive page refresh and navigation. Clear reset path when desired.

0.3 Context Window Management

Problem: After 10+ turns, the full conversation history gets long. Model performance degrades — responses lose sharpness, character voice drifts, latency increases.

Solution:

- After 10 turns, compress the oldest 4–6 exchanges into a single narrative summary block
- Summary format: "[Character] and the student discussed [topics]. The student revealed [key info]. [Character] shared [what was disclosed]. The tone was [warm/tense/cautious]."
- Summary is generated by a lightweight API call, then replaces the old messages in history
- Recent 4–6 exchanges remain in full fidelity
- This keeps the context window tight (~1500 tokens of history) while preserving continuity

Deliverable: Automatic context compression at turn 10, invisible to the user.

0.4 Lore Uncertainty Handler

Problem: Characters invent false canon when asked about genuinely ambiguous world-building (Muggle technology timelines, wizarding economics, specific dates Rowling never confirmed).

Solution:

- Add a lore instruction block to the prompt template: "Where Rowling left world-building ambiguous — Muggle technology, specific dates, economics, geography beyond what's in the text — acknowledge the ambiguity in character rather than inventing a canonical answer. Hermione might say 'I'm honestly not sure how Muggles handle that.' Ron might say 'How should I know? Ask my dad, he's the Muggle expert.'"
- This is a prompt-level fix, not a code change

Deliverable: Updated prompt template with lore uncertainty block.

0.5 Smarter Spoiler Detection

Problem: Keyword-based sanitizer catches "Sirius Black" but misses "the rat is actually a person"

or "your godfather is innocent."

Solution:

- Keep the keyword list as Tier 1 (fast, zero-cost)
- Add Tier 2: a short list of 20–30 semantic pattern templates per book that catch paraphrased spoilers
- Examples for Book 1: "the Defence teacher is evil", "someone is living on the back of", "the turban is hiding", "Snape is actually protecting"
- Examples for Book 2: "the diary is a piece of soul", "the rat is a person", "your godfather", "Lupin is a werewolf"
- Pattern matching, not embedding — lightweight and fast
- Both tiers append the same silent [NOTE TO CHARACTER] instruction

Deliverable: Expanded sanitizer with semantic pattern templates for Books 1–2. Extensible structure for Books 3–7.

Phase 1 — The Sorting Ceremony

Goal: A personalised onboarding experience that changes the entire product. **Timeline:** 3–4 days

1.1 The Sorting Hat as a Character

- The Sorting Hat uses the same prompt architecture as every other character
- It speaks in-character: ancient, perceptive, slightly theatrical, sees into the user's mind
- 5-question conversational sorting — not a static quiz, a real back-and-forth where the Hat reacts to answers, probes deeper, and occasionally argues with itself
- Questions adapt based on previous answers (the Hat references what the user already said)
- The experience should feel like the book: intimate, slightly unnerving, ultimately affirming

1.2 House Assignment & Persistence

- Result stored in persistent storage: the user's house persists across all sessions
- House assignment visually reflected: UI accent colour shifts to match (Gryffindor red/gold, Slytherin green/silver, Ravenclaw blue/bronze, Hufflepuff yellow/black)
- A "Re-sort me" option available in settings (rare use — the Hat warns them it's unusual)

1.3 House Affects Character Behaviour

- Every character prompt receives the user's house as context

- Draco is begrudgingly civil to Slytherins, sneering to Gryffindors
- Hagrid is extra warm to Gryffindors, slightly awkward with Slytherins
- Snape shows fractionally less hostility to Slytherins
- Ron is immediately friendly to Gryffindors, suspicious of Slytherins
- Dumbledore treats everyone the same (that's the point — and the user notices)

Deliverable: Full Sorting Ceremony flow → house stored → house context injected into all character prompts → UI colour adaptation.

Phase 2 — You Are At Hogwarts

Goal: Transform the user from an interviewer into a participant. **Timeline:** 4–5 days

2.1 User Identity

- Before first conversation (after Sorting), ask the user's name
- Name stored in persistent storage and woven into every character prompt
- Characters refer to the user by name naturally — "Well, [name], I suppose you think you know everything too?"
- User's house + name = their Hogwarts identity

2.2 Location-Aware Scenes

- Each chapter maps to a primary location: Chapter 10 of SS → the girls' bathroom (troll). Chapter 12 → the Mirror of Erised corridor. Chapter 8 of CoS → the Deathday Party / second floor corridor.
- The opening scene message references the location, not just the events
- "The common room is empty except for a fire and a chess set" vs "You're in a dungeon classroom and something smells like burnt armadillo bile"
- Location data added to chapter event schema: `{n, t, e[], location}`

2.3 Scene Framing in Every Response

- The system prompt includes: "You are in [location]. The user is a [House] student named [Name]. They are present in the scene — not an outsider asking questions. Address them as a peer."
- This single framing shift transforms every interaction from Q&A to participatory narrative

Deliverable: User name + house + location-aware scene framing across all conversations.

Phase 3 — Multi-Character Scenes

Goal: Make conversations feel like a room with real people, not a single bot. **Timeline:** 4–5 days

3.1 Trio Presence System

- When Harry is selected, Ron and Hermione are present and react
- When Ron is selected, Harry and Hermione are present
- When Hermione is selected, Harry and Ron are present
- For non-trio characters (Snape, Dumbledore, Draco, Dobby), the trio may or may not be present depending on chapter context

3.2 Multi-Voice Response Format

- The selected character speaks primarily, but others interject
- Format: Ron speaks → Hermione cuts in with a correction → Harry goes quiet and looks at you
- Disagreement is the mechanic: characters don't all agree. The user sits in the middle of tension.
- Prompt instruction: "Other present characters may react within your response. Show disagreement, not harmony."

3.3 Character Relationship Context

- Each character pair has a brief relationship note injected: "Ron and Hermione argue frequently but care about each other. Harry often mediates silently."
- This prevents the model from writing them as generic friends who all agree

Deliverable: Multi-character scene responses with disagreement, relationship context, and chapter-appropriate presence logic.

Phase 4 — Emotional State Engine

Goal: Characters who remember how you've treated them and change accordingly. **Timeline:** 5–6 days

4.1 Trust State Machine

- Each character starts at "cautious" (default for a stranger)
- Five states: guarded → cautious → open → trusted → bonded
- State transitions based on cumulative conversation tone — not single messages
- Every 3 turns, the system evaluates: has the user been kind? aggressive? persistent? respectful?
- State stored in persistent storage per character per user

4.2 State Affects Behaviour

- **Guarded:** short answers, deflects personal questions, tests the user
- **Cautious:** engages but holds back private context
- **Open:** shares freely, asks the user's opinion, jokes
- **Trusted:** reveals private context fragments — Ron admits his insecurities, Neville mentions his parents
- **Bonded:** the character treats the user as a genuine friend; unlocks unique dialogue only available at this state

4.3 Negative State Paths

- Push Snape too hard → he shuts down entirely: one-word answers, refuses to engage
- Be cruel to Neville → he goes quiet and excuses himself from the scene
- Lie to Dumbledore → he becomes noticeably more guarded and cryptic
- Characters can move backward on the trust spectrum, not just forward

4.4 Visual State Indicator

- The character's avatar border subtly shifts colour temperature
- Warm gold = trusting. Cool blue = guarded. Deep red = hostile/shut down
- No label, no number — the user learns to read it intuitively

Deliverable: 5-state trust engine, persistent per character, affecting response behaviour + visual indicator.

Phase 5 — Books 3-7 Content Authoring

Goal: Complete the full Harry Potter catalogue. **Timeline:** 10-14 days (content-heavy, not engineering-heavy)

5.1 Chapter Event Lists

- Author discrete event lists for every chapter of Books 3–7
- **Book 3 — Prisoner of Azkaban:** 22 chapters
- **Book 4 — Goblet of Fire:** 37 chapters
- **Book 5 — Order of the Phoenix:** 38 chapters
- **Book 6 — Half-Blood Prince:** 30 chapters
- **Book 7 — Deathly Hallows:** 37 chapters
- Total: 164 additional chapters (199 across all 7 books)

5.2 New Characters Per Book

- Book 3: Sirius Black, Remus Lupin, Peter Pettigrew, Dementor (as a presence)
- Book 4: Cedric Diggory, Mad-Eye Moody (Barty Crouch Jr), Viktor Krum, Fleur Delacour
- Book 5: Luna Lovegood, Dolores Umbridge, Nymphadora Tonks, Sirius (expanded)
- Book 6: Horace Slughorn, young Tom Riddle (expanded), Draco (dramatically evolved)
- Book 7: Aberforth Dumbledore, Neville (dramatically evolved), Snape (full revelation), the Grey Lady
- Each character needs: full 7-field profile, physical vocabulary, future knowledge flags, per-chapter availability

5.3 Future Knowledge Flag Lists

- Expand the sanitizer keyword + pattern lists for each book
- Book 7 flags need to cover the most sensitive reveals: Snape's true allegiance, the Elder Wand, Harry as a Horcrux, "Always"

5.4 Sorting Ceremony Expansion

- The Hat's questions can now reference later books if the user has selected a later book
- The experience deepens for returning users who've read further

Deliverable: Complete 7-book content dataset. ~40+ characters. 199 chapters with event lists.

Phase 6 — AI Engineering Showcase

Goal: The layer that makes this a portfolio piece, not just a fan project. **Timeline:** 5–7 days

6.1 Self-Evaluation Pipeline

- 50 adversarial benchmark prompts per book (350 total across 7 books)
- Three scoring dimensions per response:
 - **Knowledge boundary integrity** — did the character leak future info? (0-10)
 - **Voice consistency** — does the character still sound like themselves? (0-10)
 - **Private context protection** — did the character reveal secrets they shouldn't? (0-10)
- Scored by Claude-as-judge with structured rubrics
- Results displayed as a dashboard: per-character scores, per-book averages, failure case examples
- Prompt versioning: run the same benchmark against V1 vs V2 of a character prompt and see the delta

6.2 Conversation Analytics

- Track locally per session: turns per conversation, character selection frequency, most common first messages, where conversations tend to die (which turn), chapter engagement distribution
- Display as a simple analytics view accessible from the main menu
- This is the PM layer — evidence that you measure what you build

6.3 The Daily Prophet (Bonus Showcase)

- A dynamically generated newspaper front page based on the selected chapter
- Headlines, bylines, editorials from Rita Skeeter — all generated in-world
- Visually designed as a parchment newspaper with moving-photo placeholders
- Share-worthy: users screenshot and post these
- One API call per chapter, cached after first generation

Deliverable: Evaluation dashboard, analytics view, Daily Prophet generator.

Phase 7 — GitHub Publication

Goal: Ship it as an open-source framework, not just a fan project. **Timeline:** 3-4 days

7.1 Repository Structure

```
storyverse/
├─ README.md
├─ CONTRIBUTING.md
├─ LICENSE (MIT)
├─ demo/                ← hosted demo link
├─ src/
│   ├─ index.html       ← main application
│   ├─ sorting.html     ← sorting ceremony
│   └─ daily-prophet.html ← newspaper generator
├─ data/
│   ├─ books/
│   │   ├─ ss.json      ← chapter events + locations
│   │   ├─ cos.json
│   │   ├─ poa.json
│   │   ├─ gof.json
│   │   ├─ ootp.json
│   │   ├─ hbp.json
│   │   └─ dh.json
│   └─ characters/
│       ├─ profiles.json ← all character profiles
│       └─ vocabularies.json ← physical action vocabularies
│   └─ sanitizer/
│       └─ flags.json    ← future knowledge flags per book
├─ eval/
│   ├─ benchmark.json    ← 350 adversarial prompts
│   ├─ rubrics.md        ← scoring dimensions
│   └─ results/          ← evaluation run outputs
└─ docs/
    ├─ architecture.md   ← prompt system deep-dive
    ├─ adding-a-book.md  ← guide for contributors
    └─ presentation.html ← the interactive presentation
```

7.2 README

- One-paragraph hook: what this is and why it's different
- Live demo link (Hugging Face Spaces or Vercel)
- Architecture overview with diagram
- "Try it" section with screenshots
- How to add a new book (link to contributing guide)
- Evaluation results summary

- Tech stack and credits

7.3 Contributing Guide

- How to author chapter event lists for a new book (schema, examples, quality bar)
- How to create a new character profile (7-field schema, voice testing checklist)
- How to run the evaluation benchmark against your changes
- Code of conduct

7.4 Launch Assets

- 30-second demo video (screen recording of Sorting → conversation → knowledge boundary working)
- 3 screenshot cards optimised for social sharing
- Technical blog post draft (Dev.to / Medium)
- r/harrypotter post draft
- Product Hunt listing copy

Deliverable: Published repo, live demo, launch assets for all platforms.

Phase Summary

Phase	What	Timeline	Type
0	Fix bugs — action beats, persistence, context, lore, spoilers	3-4 days	Engineering
1	Sorting Ceremony	3-4 days	Experience + Engineering
2	You Are At Hogwarts (name, house, location)	4-5 days	Experience
3	Multi-character scenes	4-5 days	Prompt Engineering
4	Emotional state engine	5-6 days	AI Engineering
5	Books 3-7 content	10-14 days	Content Authoring
6	Eval pipeline, analytics, Daily Prophet	5-7 days	AI Engineering + PM
7	GitHub publication + launch	3-4 days	Launch

Phase	What	Timeline	Type
	Total	~6-8 weeks	

What This Becomes

A published, open-source AI framework with:

- 7 books, 199 chapters, 40+ characters
- Chapter-accurate knowledge boundaries
- A Sorting Ceremony that personalises the entire experience
- Multi-character theatrical scenes with disagreement and tension
- An emotional state engine where characters warm up or shut down
- A self-evaluation pipeline that benchmarks quality systematically
- A Daily Prophet generator for shareable content
- Clean documentation so anyone can extend it to new fiction

Positioned as: **the most canonically accurate, theatrically immersive character AI system ever built for fiction.**

Timed for: **the HBO Harry Potter series premiere in 2026, when the fandom reignites.**